I'd like to write a note about martingale limit theorems. The main reference for the moment is Amir's lecture notes, Pollard 2002 and Hall and Heyde 2014.

The topics are(growing):

1. Independence and Construction of Measure on Product Space.
2. Concentration Inequality from **Bartlett lecture notes**.
3. Conditional Expectation. (Pollard 2002, Amir Probability notes).
4. *Stochastic Convergence
5. *Laws of Large Numbers
6. *Central Limit Theorems
7. *Laws of Iterated Logarithm
8. Martingales and Doob's Convergence Theorems.
9. *Brownian Motion
10. *Ergodic Theory
11. Martingale inequalities. Hall and Heyde 2014.
12. Martingale Law of Large Numbers.
13. Martingale CLT.

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Chapter 1

Kernel, Independence and Conditional Expectation

1.1 Independence

Fist we define the independence.

For (finite) product space $(X \times Y, \mathcal{A} \otimes \mathcal{B})$, first what is in $\mathcal{A} \otimes \mathcal{B}$?

Kernel or Conditional Distribution. Let $(X, \mathcal{A}, \mu)$ be a measure space, a kernel from $(X, \mathcal{A})$ to $Y, \mathcal{B}$ is a family of measure $\Lambda = \{\lambda_x : x \in X\}$, such that $x \mapsto \lambda_x B$ is $\mathcal{A}$-measurable for all $B \in \mathcal{B}$.

If $\lambda_x(Y) < \infty$ for all x, then we can easily construct a measure on $\mathcal{A} \otimes \mathcal{B}$ by the iterated integral:

$$\mu^x(\lambda_x^y f(x, y)) \quad \text{for all} \quad f(x, y) \in \mathcal{M}(\mathcal{A} \otimes \mathcal{B}).$$

1.1.1 Infinite Product Spaces

1.1 Daniell and Kolmogorov. *Topological statement about when we can extend a consistent family of finite-dimensional probability measure to an infinite product space.* ◇

1.2 Conditional Expectation

Conditioning is a getaround of independence conditions.

There is a way to define conditional distribution and conditional density with the kernel. However we can start with Kolmogorov's abstract definition of conditional expectation.

We can define the conditional expectation from the projection in L^2, or from *Radon-Nikodym* derivative.

Conditional Expectation. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ be a sub-σ-algebra of $\mathcal{F}$, let $Y \in L^1(\mathcal{F})$, then define the conditional expectation Z as a $\mathcal{G}$ measurable r.v. such that:

$$E(ZG) = E(YG) \quad \text{for all} \quad G \in \mathcal{G}$$

1.2 Existence and Uniqueness of Conditional Expectation. ◊

1.3 Lemma (Radon-Nikodym). *Let m be an unsigned σ-finite measure, μ be a signed σ-finite measure, then if $\mu \ll m$ then*

$$\mu = m_f \quad \textit{for some} \quad f \in L^1(X, \mathrm{d}m)$$

1.4 Proposition (Properties of Conditional Expectation).

1.2.1 Properties of CE

1.5 Proposition (Increasing Linear with Monotone Convergence). *Let $X, Y \in L^1(\Omega, \mathcal{F})$, then we have for a sub-σ-algebra $\mathcal{H}$, we have:*

1. *Linearity of conditional expectations,*

$$E(aX + bY \mid \mathcal{H}) = aE(X \mid \mathcal{H}) + bE(Y \mid \mathcal{H})$$

2. *Monotone, if $X \leq Y$ a.s., then*

$$E(X \mid \mathcal{H}) \leq E(Y \mid \mathcal{H}) \quad a.s.$$

3. *Monotone convergence: if $X_n \uparrow X$, then $E(X_n \mid \mathcal{H}) \uparrow E(X \mid \mathcal{H})$.*

1.6 Jensen's Inequality. *If $\Phi : \mathbb{C} \to \mathbb{C}$ is convex, then for $X \in L^1$ and $\Phi(X) \in L^1$, we have*

$$\Phi(E(X \mid \mathcal{H})) \leq E(\Phi(X) \mid \mathcal{H})$$

$\diamond$

Proof. Convex functions are sup of a countable set of affine functions. $\Phi(X)$ is also measurable because for any $t \in \mathbb{R}$, $\Phi^{-1}(\leq t)$ is an interval. For each affine function, there exists a null set, outside of which the result holds, so the sup outside the union of the null sets will give the result. $\square$

Corollary 1.6.1. $\|E(X \mid \mathcal{H})\|_{L^p} \leq \|X\|_{L^p}$, *for* $p \geq 1$.

Chapter 2

Stochastic Convergence

2.1 Convergence Concepts

2.2 Martingale Convergence

2.2.1 Up-crossing Inequality and Almost Sure Convergence

2.2.2 Doob's Inequality and L^p Convergence

2.2.3 Martingale CLT

There are important *invariance principles* and *CLT*.

Techniques usually mimic the classical methods with independent r.v. But there are novel techniques, for example, Strassen's representation(cf. Skorokhod's representation for iid.)

Various sufficient conditions:

1. Asymptotic negligibility.

2. Conditional Variance: it's a bit like the Fisher information. So we need some regularity condition on that.

3. Relationship between conditional variance and U_{ni}^2.

4. Stability, L^1-convergence, mixing conditions. Häusler and Luschgy 2015. Convergence in distribution *stably* allows for norming with a random variable instead of a constant in the Slutsky's theorem.